

TorqCapture — Design Study v1

*Reframe "capture the flag" into **capture the region** — territory control the way FPS games (Call of Duty) do it, with a **playlist of modes on one setup**. This study decides the core mechanic and the mode set, and settles how it relates to the existing "Capture the Zone." Feeds a future rulebook + needs doc if the direction is picked. Same standalone-event frame as TorqFall/TorqSiege.*

The reframe: flag-run → territory-hold

The current **Capture the Zone** is capture-the-flag / payload: grab a neutral object, drive it to your base, most deliveries win. It's a **fetch quest** — a solo errand, low contact, one clear "right line."

TorqCapture is the FPS objective family instead — **hold ground, don't carry things**. You occupy a glowing region; while your team owns it, it ticks points; the other team drives in to take it back. It's a **fight over map control**, continuous and positional — and it naturally supports the thing Yash wants: **multiple game modes on the same floor**, swapped like a COD playlist.

The core mechanic (the crux — cars can't shoot)

Every game has one unknown to solve; here it's "**how does a speed-limited RC car capture and contest a zone?**" The answer:

- **Occupy = hold.** A zone is a marked ring (~1.5–2 m). A car of your colour physically inside it holds it → it ticks points for your team.
- **Deny = shove.** A rival drives in to contest; you **body-check them out of the ring**. The "weapon" is the bumper — this is **TorqSumo's push mechanic aimed at an objective**. Same speed cap + same foam bump-rings as TorqFall keep it safe.
- **Scored electronically (LOCKED):** each car carries a **team-coloured tag/beacon**; each zone has a **short-range reader**. The system knows, live, **which cars — and how many per team — are inside each zone**.
- **Ownership = the count.** The team with **more cars in the zone** controls it; equal (or empty) = **contested**, the timer pauses. So it's literally "*who has more cars on the ground here*" — pile in, shove theirs out, hold the number.
- **Capture = hold the majority for a timer** (e.g. 5 s continuous) → the zone flips to your colour and starts ticking points. Lose the majority and the enemy's flip-timer starts.
- **Tech is the scorer** (a marshal is backup for disputes) — the glowing zone changes colour on capture and auto-updates the big screen. Engineering in [TorqCapture — Zone-Capture Tech](#) (NFC is too short-range — this is RFID / BLE / vision territory).

So TorqCapture = TorqSumo (push) + an objective (hold) + COD variety (modes). It reuses the fleet, the bump-rings, the astroturf, and the shove mechanic we already understand — very little new to build.

The playlist — COD modes → TorqCapture modes

One floor, swap the rules + zone layout in minutes. This is the headline feature.

COD mode	TorqCapture mode	How it runs with RC cars	Why it's good
Domination	Domination (default)	3 zones A / B / C across the floor; hold a zone = it ticks points/sec; own 2 of 3 = the tug-of-war	The readable classic — map splits into a front line; best base mode
Hardpoint	Hotzone	ONE lit zone; hold it; every 60–90 s it relocates to a new spot → whole pack scrambles across the floor	Constant repositioning = the scramble is the money content
King of the Hill / Control	The Hill	ONE central zone, hold longest; simplest rules	The quick round + the showpiece final
Headquarters	Siege Point (advanced)	Capture a zone; once held it locks for N s (can't be retaken) → assault/defend rhythm; then it despawns + a new one spawns	Adds a defend phase; save for when the crowd knows the game
Kill Confirmed / payload	Courier (legacy)	The old Capture-the-Zone flag-run kept as ONE mode in the playlist — grab a neutral core, deliver to your zone	Keeps the fetch variant alive as flavour, not the whole game

A **TorqCapture event** = a **mode rotation** (a "playlist"): Domination heats → Hotzone → a Hill final. Like a real COD session, and it keeps a full standalone day varied without new hardware.

Beginner fun

Very high. "Drive into the glowing circle and stay there while your teammate blocks the door" is instant — no skill wall. **Team colours = built-in narrative + cheering**. The combat is shoving, which everyone can do and nobody feels bad losing. Chaos in a small zone hides skill, exactly like TorqFall. Arguably the most *social* game we have — it's the only true **team-vs-team territory** game in the set.

Content

The zones **flip colour on the overhead cam** as control changes — a live, legible map of who's winning (great for the big screen + reels). The **Hotzone relocation scramble** and the last-second contested flip are the money frames. Team colours pop on camera. Commentary writes itself ("Blue's lost B! they're scrambling!").

Relationship to "Capture the Zone" (a real decision)

TorqCapture is the **stronger evolution** of Capture the Zone — a fight instead of a fetch, and it's *moded*. Recommendation: **TorqCapture supersedes Capture the Zone as the "capture" game**, and the old flag-run survives as the **Courier** mode inside it. Capture the Zone was graded **Best** but never built out (Notion page only) — so nothing is lost, it's absorbed and upgraded. (Alternative: keep both as separate games — but that's two overlapping "capture" games competing for one slot; not worth it.)

Where it slots

Its own **standalone event** (a TorqCapture playlist tournament — most content-first, most team-driven) OR a station inside a bigger day. As a standalone it reuses ~everything from the TorqFall base kit (fleet, turf, barriers, show layer, crew) + a small zone kit — likely the **cheapest new event to stand up** after TorqFall, because there's almost nothing novel to fabricate.

The one thing to prototype (its "collapse test")

Can the zone reader reliably detect + count the cars inside a ~1.5–2 m ring — and only those — while they shove and overlap? That's the gate (same role the collapse test plays for TorqSiege). NFC is out (cm range); the real candidates are **UHF RFID** (a tag per car + a zone antenna), **BLE/RSSI beacons** (ESP32), or a **computer-vision overhead cam** (coloured/marker tops — spatially exact, auto-scores, doubles as the content feed). Cheap bench test + the recommended build + India-available modules live in [TorqCapture — Zone-Capture Tech](#). Tune: ring size, reader range/confinement, the capture timer, and whether a raised foam ring helps cars hold position.

Decisions locked

1. **Absorb** — TorqCapture **replaces** Capture the Zone; the old flag-run survives as the **Courier** mode. Capture the Zone → superseded in the index/Notion.
2. **Standalone game** — built as its own game in the directory (like every game); whether it runs as a club/mixed day or solo event is decided later per plan. This doc set treats it standalone.
3. **Electronic scoring** — build the tag + reader zone tech (it draws a crowd); marshal is backup, not the scorer. Speced in [TorqCapture — Zone-Capture Tech](#).

Doc set: [TorqCapture — Rulebook](#) (5 modes, zone specs, count/hold capture rules, match flow, safety, content), [TorqCapture — Zone-Capture Tech](#) (RFID/BLE/vision analysis + recommended build + BOM + bench test — the prototype gate), [TorqCapture — What The Event Needs](#), [TorqCapture — Full Run Plan](#), and the Notion page + [Game Ideas - Graded](#) index absorbing Capture the Zone. Pairs with [Event Ops Standards \(all games\)](#) + the shared fleet/protection kit.